**REVISION WORKSHEET**

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

# AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

## Practice Questions and Worked Answers

**Grade 11 Advanced | Module 8 Lesson 1 | Describing Rotational Motion**

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $\theta = s/r$; $\omega_{avg} = \Delta\theta/\Delta t$; $\alpha_{avg} = \Delta\omega/\Delta t$; $v = r\omega$; $a_t = r\alpha$; $\omega = 2\pi f$; 1 revolution = $2\pi$ rad.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 3–6.

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**P01. One complete revolution equals:**

A. π radians  B. 2π radians
C. 180 radians  D. 1 radian

**Answer**

B. 2π radians.

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**P02. Points on the same rigid wheel share the same:**

A. Tangential speed at every radius  B. Distance travelled in every rotation
C. Angular velocity  D. Tangential displacement magnitude

**Answer**

C. Angular velocity.

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**P03. For fixed angular velocity, tangential speed is:**

A. Inversely proportional to radius  B. Independent of radius always
C. Equal to radius without a time factor  D. Proportional to radius

**Answer**

D. Proportional to radius.

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**P04. Angular acceleration is the rate of change of:**

A. Angular velocity  B. Radius
C. Mass  D. Moment of inertia only

**Answer**

A. Angular velocity.

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**P05. A radian is related to arc length s and radius r by:**

A. $\theta = sr$  B. $\theta = s/r$
C. $\theta = r/s$ for all angles  D. $\theta = s + r$

**Answer**

B. $\theta = s/r$.

**P06. Convert 90° to radians.**

A. 3.142 rad  B. 0.7854 rad
C. 1.571 rad  D. -1.571 rad

**Answer**

C. 1.571 rad. $\theta$ = degrees × $\pi/180 = 90\pi/180 = 1.571$ rad.

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**P07. A wheel turns through 8 rad in 4 s. Calculate its average angular velocity.**

A. 4 rad/s  B. 1 rad/s
C. -2 rad/s  D. 2 rad/s

**Answer**

D. 2 rad/s. $\omega_{avg} = \Delta\theta/\Delta t = 8/4 = 2$ rad/s.

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**P08. A point is 0.2 m from a rotating axle with angular velocity 5.0 rad/s. Calculate its tangential speed.**

A. 1 m/s  B. 2 m/s
C. 0.5 m/s  D. -1 m/s

**Answer**

A. 1 m/s. $v = r\omega = 0.2 \times 5.0 = 1$ m/s.

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**P09. A wheel's angular velocity changes from 2 to 14 rad/s in 6 s. Find its average angular acceleration.**

A. 4 rad/s²  B. 2 rad/s²
C. 1 rad/s²  D. -2 rad/s²

**Answer**

B. 2 rad/s². $\alpha = (\omega_f - \omega_i)/\Delta t = (14 - 2)/6 = 2$ rad/s².

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**P10. A wheel rotates at frequency 4 Hz. Calculate its angular velocity.**

A. 50.27 rad/s  B. 12.57 rad/s
C. 25.13 rad/s  D. -25.13 rad/s

**Answer**

C. 25.13 rad/s. $\omega = 2\pi f = 2\pi \times 4 = 25.13$ rad/s.

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**P11. Define angular displacement.**

**Answer**

The signed change in angular position about a specified rotation axis, usually measured in radians.

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**P12. Define average angular velocity.**

**Answer**

Angular displacement divided by elapsed time: $\omega_{avg} = \Delta\theta/\Delta t$.

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**P13. Define average angular acceleration.**

**Answer**

The change in angular velocity divided by elapsed time: $\alpha_{avg} = \Delta\omega/\Delta t$.

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**P14. How are tangential and angular acceleration related for a fixed radius?**

**Answer**

The tangential component is $a_t = r\alpha$. It is distinct from radial centripetal acceleration.

**P15. Explain why radians should be used in $v = r\omega$.**

**Answer**

A radian is defined by arc length divided by radius, so using radians makes the arc-length and tangential-speed relations hold without an extra conversion factor.

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**P16. Convert 135° to radians.**

**Answer**

$\theta$ = degrees × $\pi/180 = 135\pi/180 = 2.356$ rad.

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**P17. Convert 180° to radians.**

**Answer**

$\theta$ = degrees × $\pi/180 = 180\pi/180 = 3.142$ rad.

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**P18. A wheel turns through 12 rad in 5 s. Calculate its average angular velocity.**

**Answer**

$\omega_{avg} = \Delta\theta/\Delta t = 12/5 = 2.4$ rad/s.

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**P19. A wheel turns through 16 rad in 6 s. Calculate its average angular velocity.**

**Answer**

$\omega_{avg} = \Delta\theta/\Delta t = 16/6 = 2.667$ rad/s.

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**P20. A point is 0.3 m from a rotating axle with angular velocity 5.0 rad/s. Calculate its tangential speed.**

**Answer**

$v = r\omega = 0.3 \times 5.0 = 1.5$ m/s.

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**P21. A point is 0.4 m from a rotating axle with angular velocity 5.0 rad/s. Calculate its tangential speed.**

**Answer**

$v = r\omega = 0.4 \times 5.0 = 2$ m/s.

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**P22. A wheel's angular velocity changes from 3 to 21 rad/s in 9 s. Find its average angular acceleration.**

**Answer**

$\alpha = (\omega f - \omega i)/\Delta t = (21 - 3)/9 = 2$ rad/s².

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**P23. A wheel's angular velocity changes from 4 to 28 rad/s in 12 s. Find its average angular acceleration.**

**Answer**

$\alpha = (\omega f - \omega i)/\Delta t = (28 - 4)/12 = 2$ rad/s².

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**P24. A wheel rotates at frequency 6 Hz. Calculate its angular velocity.**

**Answer**

$\omega = 2\pi f = 2\pi \times 6 = 37.7$ rad/s.

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**P25. A wheel rotates at frequency 8 Hz. Calculate its angular velocity.**

**Answer**

$\omega = 2\pi f = 2\pi \times 8 = 50.27$ rad/s.

**P26. Two points on a rigid wheel are at radii r and 2r. Compare their angular velocities and tangential speeds.**

**Answer**

Their angular velocities are equal. Since $v = r\omega$, the outer point has twice the tangential speed.

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**P27. A wheel rotates at constant nonzero angular velocity. Is every point's linear velocity constant? Explain.**

**Answer**

No. Its tangential speed can be constant, but the direction of linear velocity changes around the circle, so it has radial acceleration.

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**P28. A disk turns 5 revolutions in 2.0 s at a steady rate. Find angular velocity and the speed of a point 0.10 m from the axis.**

**Answer**

$\Delta\theta = 10\pi$ rad; $\omega = 10\pi/2.0 = 5\pi \approx 15.7$ rad/s. $v = 0.10\omega \approx 1.57$ m/s.

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**P29. Why do different points on Earth travel different distances in a day although they complete the same rotation angle?**

**Answer**

The distance to Earth's rotation axis varies with latitude. Arc length $s = r\theta$ depends on radius, while $\theta$ is shared for a rigid rotation.

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**P30. Design a simple way to measure the angular velocity of a slow wheel.**

**Answer**

Mark one point, count a known number of revolutions during a timed interval and calculate $\omega_{avg} = 2\pi N/\Delta t$. Repeat and average, keeping the rotation rate steady if possible.